

Tab 1



Project Title - 1

AI SQL Data Analyst Agent (CSV → SQL → Insights)



Objective

Build an AI agent that:

1. Takes a **CSV file**
 2. Converts it into a **SQL database**
 3. Accepts **natural language queries**
 4. Converts them into **SQL queries**
 5. Executes them
 6. Returns:
 - Answer
 - SQL query
 - Visualization
-



System Architecture

User Input (CSV + Question)



Data Loader (Pandas)



SQLite Database



LangChain SQL Agent



Groq LLM (Llama 3)



SQL Query Generation

↓
Execution Engine
↓
Final Answer + Chart

Tech Stack

- **Frontend:** Streamlit
- **LLM:** Groq (Llama 3 / Mixtral)
- **Framework:** LangChain
- **Database:** SQLite
- **Data Handling:** Pandas
- **Visualization:** Matplotlib / Plotly

Tab 2

Project Title - 2

AI News Auto-Blogger & Email Automation using n8n + GROQ LLM

Objective

Build an automated workflow that:

1. Fetches **latest AI news** using **Serpapi**
2. Uses an LLM (Groq/OpenAI) to:
 - Summarize news
 - Generate a **blog article**
3. Formats it into:
 - Blog post
4. Automatically sends it via email

👉 End Goal:

A **fully automated AI content engine** (daily)

Workflow Architecture (n8n)



